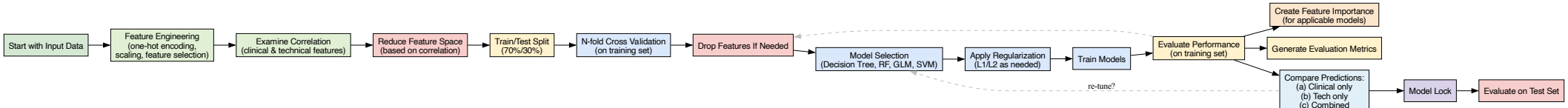
Start with Input Data
Feature Engineering (one-hot encoding, scaling, feature selection)
Examine Correlation (clinical & technical features)
Reduce Feature Space (based on correlation)
Train/Test Split (70%/30%)
N-fold Cross Validation (on training set)
Drop Features If Needed
Model Selection (Decision Tree, RF, GLM, SVM)
Apply Regularization (L1/L2 as needed)
Train Models
Evaluate Performance (on training set)
Create Feature Importance (for applicable models)
Generate Evaluation Metrics
Compare Predictions: (a) Clinical only (b) Tech only (c) Combined
re-tune?
Model Lock
Evaluate on Test Set